

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT2 — VISUALIZATION EVALUATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Evaluation Exercise
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate the appropriateness and design choices of relationship-visualization techniques (scatterplots, correlograms, PCA, bubble/hexbin plots, and network graphs) and justify improvements.(BL5)</i>		
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Section / Batch:		Date:	22.08.2026

Code of Conduct

I Priyanka.G/192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: evaluate the specific strength or weakness, and justify your judgment.
- Where a question asks for a recommendation, propose a specific alternative — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION EVALUATION EXERCISE

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5 Correlation vs. Causation Scatterplot Interpretation (5 Marks)

Data: A scatterplot shows a strong visual pattern, and the analyst concludes that X directly causes Y based solely on this chart.

Task: *Evaluate this conclusion and identify what additional evidence would be needed to support it.*

Evaluation Plan (Key Elements):

Principle Evaluated	Correlation does not, by itself, establish causation
Chart Shown	Scatterplot with fitted trend line strong positive linear pattern
Issue	Conflating a strong visual association with a proven causal effect

Evaluation & Recommendation:

The analyst's conclusion that X directly causes Y is not justified by the scatterplot alone, no matter how strong or clean the linear pattern appears. A scatterplot even one showing a near-perfect line, as in the constructed chart (Figure 5) demonstrates only statistical association between two variables. It cannot, by itself, distinguish between several very different underlying explanations that could all produce an identical pattern: X could genuinely cause Y; Y could actually cause X (reverse causation); both X and Y could be driven independently by some third, unmeasured confounding variable; or the pattern could simply be a coincidence within this particular sample.

Because all of these different underlying realities can generate the exact same scatterplot, a reader who accepted the analyst's causal claim on this evidence alone could be misled into intervening on X expecting a change in Y, only to find no effect precisely the risk of confusing correlation with causation.

If, for example, this scatterplot represented a company's advertising spend (X) against sales revenue (Y), acting on the unproven causal claim could mean increasing advertising spend expecting a proportional revenue increase, when in reality both variables might simply be rising together due to a third factor such as overall seasonal demand or economic growth in which case additional spend would not produce the assumed return, and the decision would have been made on a false premise the chart alone could never have distinguished from a true one.

Additional evidence needed to support a genuine causal claim includes:

- Temporal precedence evidence that changes in X reliably occur before corresponding changes in Y, since a cause must precede its effect in time.
- A controlled experiment or intervention directly manipulating X while holding other factors constant, then observing whether Y responds; this is the standard method for establishing causation rather than mere association.
- Control for confounding variables statistically ruling out plausible third factors that could independently produce both X and Y without any direct link between them.
- A plausible causal mechanism a substantive, domain-specific explanation for how X could physically or logically bring about a change in Y, ideally supported by prior research or established theory.

It is worth noting that the scatterplot itself remains the correct chart choice for revealing the association in the first place; the flaw is not in constructing the chart, but in the causal conclusion drawn from it a well-built scatterplot is the right first step, it simply cannot carry the full evidentiary weight of a causal claim on its own.

This distinction matters well beyond this single scenario: many well-known spurious correlations (for example, statistical curiosities linking unrelated national statistics) demonstrate that even a visually flawless linear pattern with a high correlation coefficient can arise from coincidence or shared trends over time rather than any genuine causal link, which is precisely why data-visualization practice treats a strong scatterplot pattern as the start of an investigation into causation, never as its conclusion.

Supporting Chart:

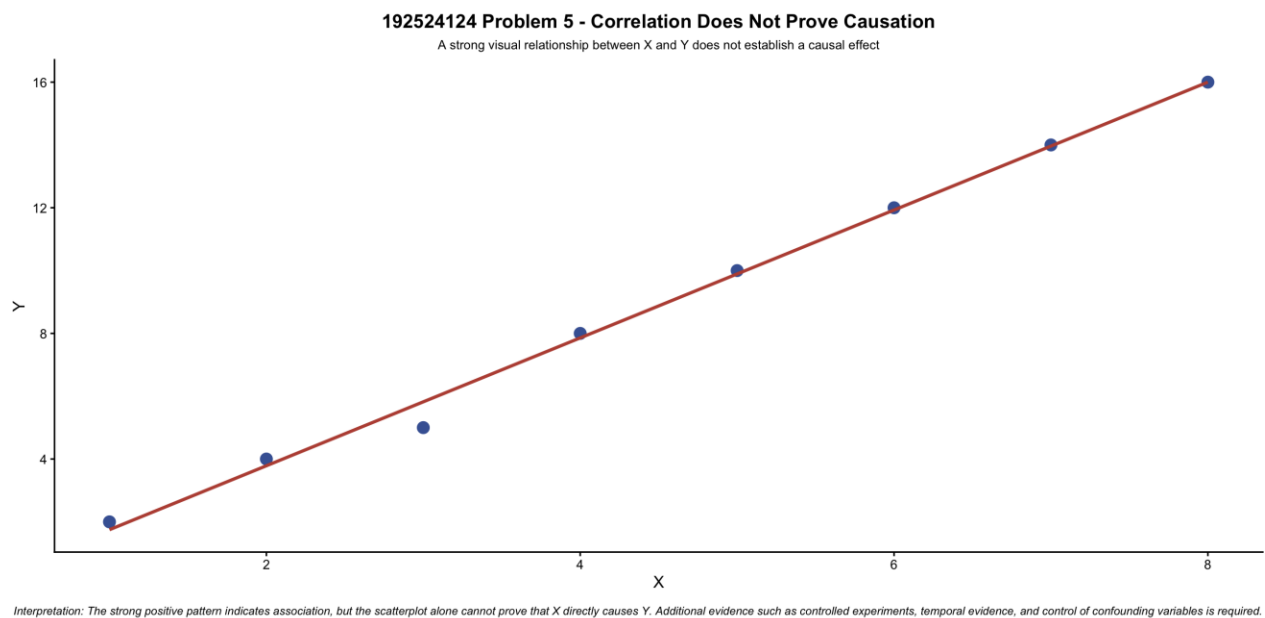


Figure 5: Scatterplot Strong Association Does Not Establish Causation

Problem 6 Correlogram for Only 2 Variables (5 Marks)

Given Data:

Pair	Correlation (r)
Variable X Variable Y	0.85

Task: Evaluate whether a correlogram is the right tool for comparing just 2 variables.

Evaluation Plan (Key Elements):

Principle Evaluated	Chart choice must match the number of variables being compared
Chart Shown	3×3-style correlogram grid used for just 2 variables
Issue	A correlogram's grid-comparison value disappears when there is only one pairwise relationship to show

Evaluation & Recommendation:

A correlogram is not the right tool for comparing only 2 variables. A correlogram's core value comes from letting a viewer compare many pairwise relationships at once, using colour to rank several cells' strength simultaneously across a full grid. With only 2 variables, there is exactly one unique off-diagonal relationship to display the matrix is symmetric and the diagonal cells are trivially 1.00 by definition so the entire 'grid comparison' value proposition of a correlogram disappears, and the chart reduces to a single large coloured rectangle conveying one number (Figure 6), which is an inefficient use of chart space and visual complexity for a single fact.

A simple correlation coefficient stated directly ($r = 0.85$) or a scatterplot of the two variables plotted against each other would each communicate the same information far more efficiently and, in the scatterplot's case, more completely. The coefficient alone gives the precise value directly, with no chart needed at all; a scatterplot additionally reveals the actual shape of the relationship whether it is linear or curved, whether outliers are pulling the correlation up or down, whether sub-groups exist within the data none of which a single colour-coded cell can show, since the cell only encodes one summary number and discards everything about the underlying pattern's shape.

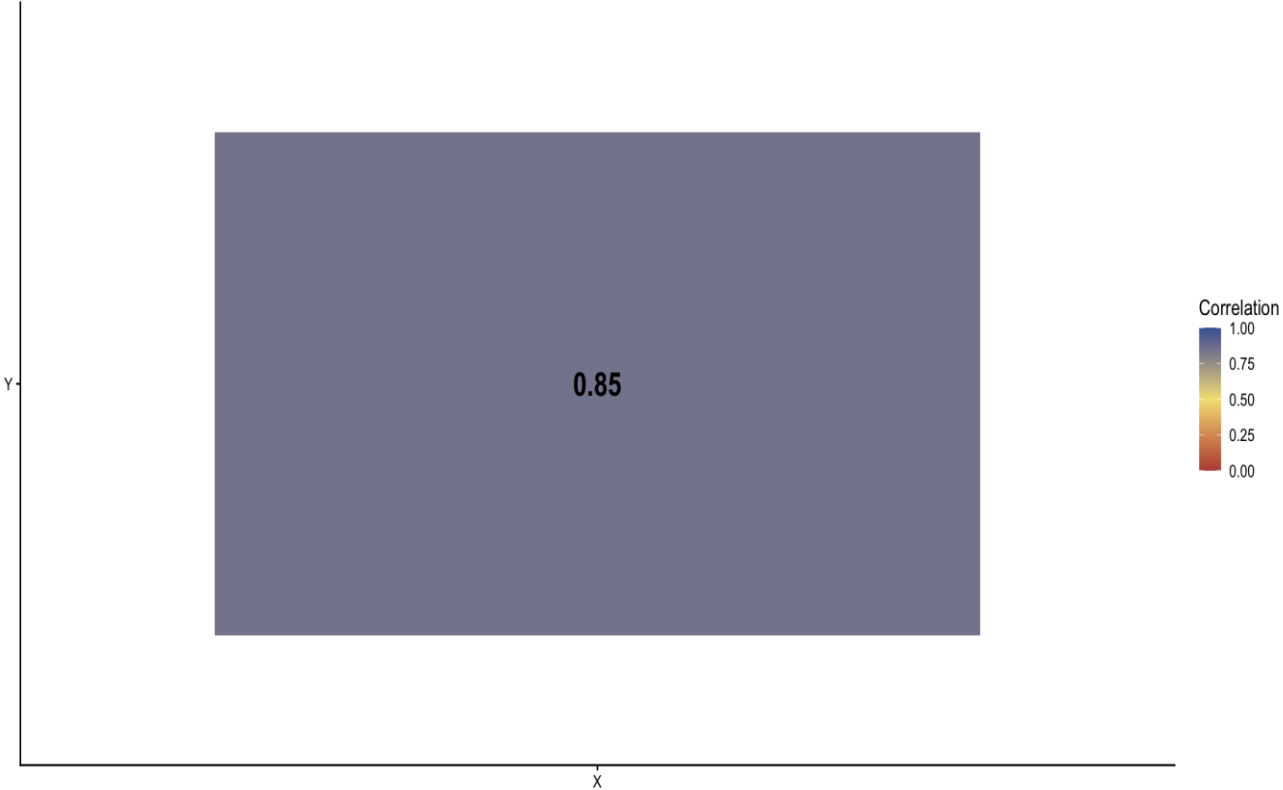
This evaluation generalises: a correlogram earns its place specifically once there are three or more variables, where its grid layout provides genuine comparative value by letting a reader scan many pairs at once and immediately spot the strongest and weakest relationships by colour exactly the comparison performed across Height, Weight, and Age in the CO4-AT1 correlogram exercise. For exactly 2 variables, that comparative advantage does not exist, and reporting the coefficient directly (optionally alongside a scatterplot) is the clearer, more appropriate choice.

More broadly, this illustrates a general principle in chart selection: a chart type's justification comes from the specific comparative or structural advantage it provides over simpler alternatives, not merely from its ability to technically display the data. A correlogram can render a 1-cell grid without error, exactly as a network graph can render a linear chain without error (see Problem 24), but 'can be rendered' is a much lower bar than 'is the most effective choice' the latter is what an evaluation of appropriateness must actually assess.

Supporting Chart:

192524124 Problem 6 - Two-Variable Correlogram

A single correlation cell provides limited comparison value



Interpretation: A correlogram is more useful when comparing several variables. With only two variables, a simple correlation coefficient or scatterplot is clearer and more informative.

Figure 6: Two-Variable Correlogram A Single Cell With Limited Comparative Value

Problem 12 PCA Variance Retention for a 2D Visualization (5 Marks)

Given Data:

Component	PC1	PC2	PC3	PC4
Variance Explained	40%	30%	20%	10%

Task: Evaluate whether this decision is justified, and what you would recommend instead.

Evaluation Plan (Key Elements):

Principle Evaluated	Retained components must both satisfy the required output dimensionality and preserve sufficient variance
Chart Shown	Bar chart of variance explained by each principal component
Issue	PC1 alone yields only 40% of variance and cannot produce a genuinely 2D plot

Evaluation & Recommendation:

The decision to keep only PC1 is not justified, for two separable reasons made visible in the constructed variance chart (Figure 12).

First, a variance-retention problem: PC1 alone explains only 40% of the total variance across all four components ($40\% + 30\% + 20\% + 10\% = 100\%$), meaning 60% of the dataset's total variability would be discarded entirely if only PC1 were kept. Any two observations that differ mainly along PC2, PC3, or PC4 collectively 60% of the dataset's structure would appear artificially similar, or even identical, once projected onto PC1 alone.

Second, and more fundamentally, a dimensionality mismatch: the stated requirement is specifically for a 2D visualization, but retaining only PC1 yields a single number per observation a 1D projection, not the two coordinates a 2D scatterplot needs. PC1 alone cannot even produce the requested 2D plot; every point would collapse onto a single axis or line, with no second dimension available at all.

In a practical setting for example, using PCA to visually separate customer segments or product clusters collapsing to PC1 alone would compress every observation onto one axis, so two genuinely distinct clusters that happen to overlap along PC1 but separate cleanly along PC2 would appear as a single indistinguishable group, actively hiding the very structure a 2D visualization was meant to reveal.

Recommendation: retain PC1 and PC2 together. This resolves both problems simultaneously it directly satisfies the 2D requirement, since two components supply exactly the two coordinates a 2D plot needs, and it captures $40\% + 30\% = 70\%$ of the total variance, nearly double what PC1 retains alone, which is a far more defensible basis for a visualization meant to represent the dataset's real structure. If 70% is still judged insufficient for a specific downstream use, a genuinely 3D visualization retaining PC1–PC3 (90% of variance) should be considered instead of compressing further into 1D.

As a general rule for dimensionality reduction, the number of components retained should be driven first by the output format's requirements (a 2D plot needs exactly 2 components, a 3D plot needs 3) and only then optimised for variance within that constraint choosing components purely by an arbitrary variance threshold without first checking they satisfy the visualization's basic dimensional requirement, as happened here, is the more fundamental

error, since no amount of additional variance from a single component can substitute for the missing second axis a 2D plot structurally requires.

Supporting Chart:

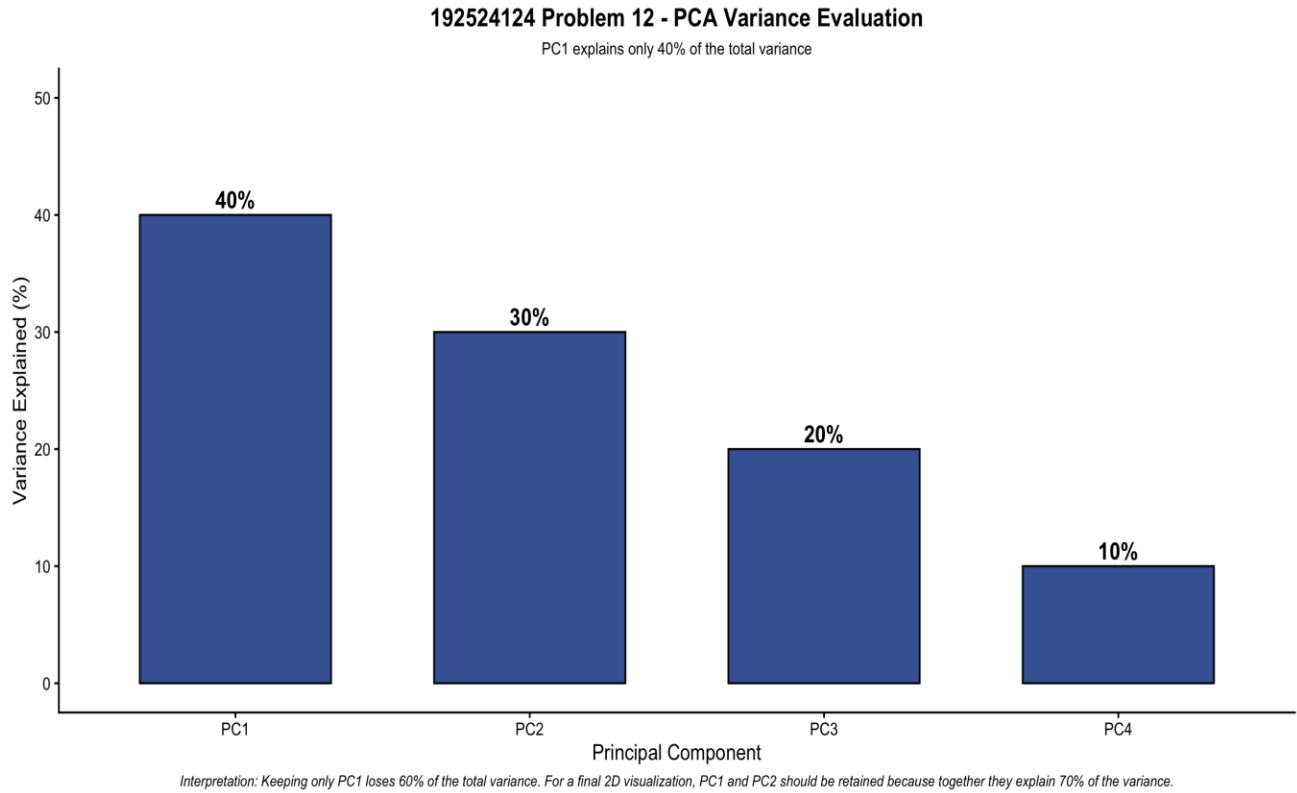


Figure 12: PCA Variance Explained by Component PC1 Alone Insufficient for 2D

Problem 18 Hexbin Binning for a Small (50-Point) Dataset (5 Marks)

Data: A hexbin plot is used to visualize a dataset containing only 50 data points.

Task: *Evaluate whether hexbin binning was necessary or appropriate for this dataset size.*

Evaluation Plan (Key Elements):

Principle Evaluated	Aggregation techniques should match the actual scale of the data
Chart Shown	Scatterplot with trend line the appropriate alternative to hexbin binning here
Issue	Hexbin binning exists to solve overplotting, which does not occur at $n = 50$

Evaluation & Recommendation:

Hexbin binning was not necessary or appropriate for this dataset. Hexbin (and other 2D-binning) techniques exist specifically to solve overplotting the situation where so many individual points are drawn that they visually merge and obscure one another, as occurs with datasets in the tens of thousands or more (for example, the 100,000-point transaction scatterplot examined in the CO3-AT3 module). With only 50 data points, this problem simply does not arise: even drawn at full opacity on a standard-sized canvas, 50 points comfortably fit as visually separable markers with no realistic risk of overlapping into an indistinguishable mass.

Applying hexbin binning here actively removes information rather than clarifying it. Binning aggregates nearby points into a single hexagonal cell and reports only a count or density per cell, discarding the ability to see each individual observation's exact position—a cost that buys nothing at $n = 50$, since clutter was never a real risk in the first place. The regular scatterplot shown in the constructed chart (Figure 18) instead preserves every individual point at its precise (X, Y) location, letting a viewer additionally see the trend's exact shape along with any outliers, gaps, or local irregularities at the level of individual observations detail a hexbin's aggregated cells would blur together.

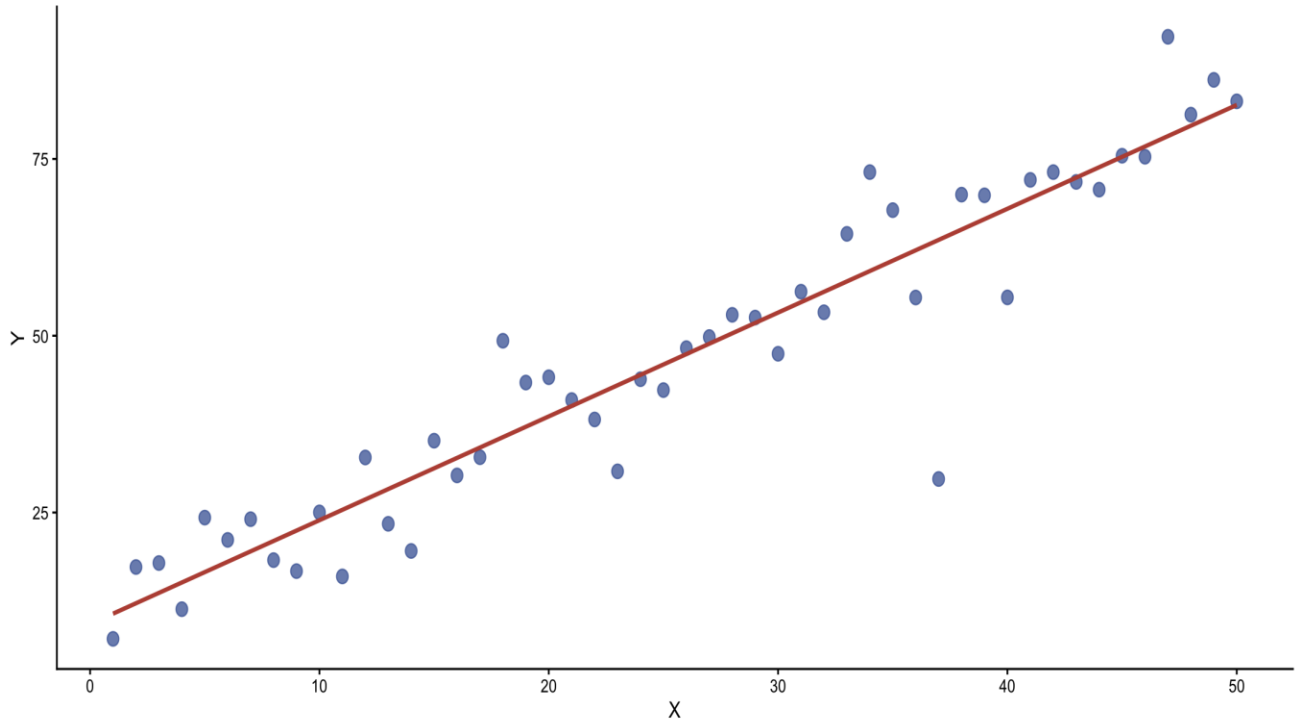
As a general guideline, hexbin or other 2D-binning techniques become genuinely useful once point counts reach roughly the thousands or tens of thousands, where a standard scatterplot would start to suffer real overplotting. For a 50-point dataset, a standard scatterplot—as used in the redesigned chart—is clearer, more transparent, and preserves strictly more information than the hexbin alternative.

This case is instructive precisely because it is the mirror image of the overplotting scenario examined elsewhere in this course: there, the fix was to move from individual points toward aggregation because there were too many marks to render distinctly; here, the correct move is the reverse—moving from aggregation back toward individual points because there were never too many marks to begin with. The lesson in both directions is the same: the appropriate level of aggregation should be dictated by the actual point count relative to the rendering scale, not applied as a default technique regardless of dataset size.

Supporting Chart:

192524124 Problem 18 - Hexbin Evaluation

Only 50 observations: a scatterplot is clearer than hexbin binning



Interpretation: With only 50 data points, overplotting is limited and individual observations can be viewed clearly. Hexbin binning is unnecessary; a standard scatterplot is more transparent.

Figure 18: Scatterplot of 50 Points Individual Observations Remain Clearly Visible

Problem 24 Network Graph for a Simple Linear Process (5 Marks)

Data: A team uses a network graph to show a simple linear process (Step 1 → Step 2 → Step 3 → Step 4) with no branching or multiple connections.

Task: *Evaluate whether a network graph was the appropriate chart type for this process.*

Evaluation Plan (Key Elements):

Principle Evaluated	Chart type must match the actual structure of the data, not just its ability to render it
Chart Shown	Simple flowchart the appropriate alternative to a general network graph here
Issue	A network graph's relational-structure strength is wasted on a single, unbranching chain

Evaluation & Recommendation:

A network graph was not the appropriate chart type for this process. A network (node-link) graph's core strength is revealing relational structure that isn't inherently ordered which nodes connect to which, whether there are multiple paths, cycles, hubs, or disconnected sub-groups, exactly the kind of structure examined in the CO4-AT1 researcher-collaboration network. None of that structural complexity exists in a strict linear sequence Step 1 → Step 2 → Step 3 → Step 4: each step connects to exactly one predecessor and one successor, with no branches, no cycles, and no alternate paths the entire 'network' is simply a single chain.

Rendering a strict chain as a general-purpose network graph is technically correct for the four steps and three connecting edges would display accurately, but it adds unnecessary visual and interpretive overhead. Network graphs typically use force-directed or otherwise non-linear layouts that can place nodes in visually arbitrary positions unless specifically constrained, requiring a viewer to trace connecting lines to reconstruct an order that a simpler chart could have shown directly through reading position alone.

A flowchart or simple process diagram, as shown in the constructed alternative (Figure 24), is the more appropriate choice: sequential boxes connected by directional arrows read naturally left-to-right, directly encoding the process order through position and arrow direction, without requiring a viewer to trace a general graph layout to recover a sequence that was linear all along. Network graphs should be reserved for genuinely relational data involving real branching, multiple connections per node, or the possibility of disconnected components none of which is present in this four-step linear process.

This evaluation pairs naturally with the CO4-AT1 researcher-collaboration network, where a network graph was the correct choice precisely because the data had genuine branching structure (a triangle of mutual collaborators alongside a separate three-person chain) that a linear chart could not represent. Placing the two cases side by side makes the underlying selection principle explicit: use a network graph when the data's connections are the object of interest and their pattern is not fully known in advance; use a flowchart when the connections are already known to form a single, predetermined sequence.

Supporting Chart:

192524124 Problem 24 - Network Graph Evaluation

A network graph is unnecessary for a simple linear process



Interpretation: A network graph is not the most appropriate chart type for this process because the process contains only a simple linear sequence: **Step 1** → **Step 2** → **Step 3** → **Step 4**. There are no branches, cycles, or multiple connections. A simple flowchart or process diagram would be more appropriate because it communicates the sequential order clearly and with less visual complexity.

Figure 24: Flowchart The Appropriate Alternative to a Network Graph for a Linear Process